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SACOS-PLATE: A new thermal-hydraulic subchannel analysis code for plate type fuel assemblies

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ABSTRACT

Over the past few decades, plate-type fuel elements have been widely used in advanced research reactors, small modular pressurized water reactors, production reactors, etc., with the advantages of compact structure arrangement, good heat transfer characteristics and high fuel consumption. Currently, a thermal–hydraulic subchannel analysis code, namely SACOS-PLATE code, was newly designed to satisfy the calculation of different types of plate fuel elements under various steady-state operating conditions. Based on the original SACOS code, this updated version modifies the flow distribution model, the plate heat conduction model, the heat distribution model as well as the flow heat transfer model, and then realizes the simulation of the special thermal–hydraulic phenomena of the fuel plate and the rectangular flow channel. In this present study, the RA-6 and Bettis experimental data was used to validate the code, and the simulation calculation was performed on the JRR-3 research reactor, which effectively proved the good computing capability of SACOS-PLATE code.

1. Introduction

Compared with the traditional rod-shaped fuel element, the plate fuel element has many advantages, such as high power-to-volume ratio, compact structure, great heat transfer characteristics and high fuel consumption. It shows superior performance in the design of nuclear reactor thermo-hydraulic system and has been widely used in advanced research reactors (Gong et al., 2015), small modular pressurized water reactors (Kim et al., 2014), production reactors (Taleyarkhana, 1992) and new nuclear-powered ship reactors (Xia et al., 2023) etc. More significantly, thermal–hydraulic analysis of plate-type fuel elements can provide theoretical guidance to the optimization design and practical application of related core design. At present, experimental research on plate fuel elements have been extensively carried out worldwide. However, due to economic and safety constraints, most experiments are performed on a single fuel element or a single flow channel (Zhang et al., 2021), and it is difficult to conduct a complete experimental study on the entire core, especially for some high-risk transient accidents. Given these facts, code simulation is often a more flexible and feasible solution than experimental research. A validated code can predict thermal parameters such as fuel plate temperature, flow heat transfer modes, critical heat flux, and others, effectively ensuring the safety and stability of reactor operation under normal conditions and providing a timely response to accident conditions.

At present, there are four main codes that can be used for thermal–hydraulic analysis of plate-type fuel element reactors (Su et al., 2016):

1. Computational fluid dynamics (CFD) codes. Xia et al. (Xia et al., 2021) simulated the blockage accident of the IAEA 10 MW MTR research reactor by STAR-CCM +. While Song (Song et al., 2013) used Fluent to study the blocking of the IAEA 10 MW MTR research reactor.
2. Dedicated thermal hydraulic code. The PARET developed by the Argonne National Laboratory (ANL) is one of the most widely used codes (Woodruff and Smith, 2001). A large number of scholars (Badry and Gaheen, 2012; Vadi and Sepanloo, 2016; Woodruff, 1984) have used PARET for steady-state and transient thermal–hydraulic analysis. Bousbia-Salah and Hamidouche (2005) developed the RETRAC-PC code to analyze the transient behavior of the

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Nomenclature

Greek symbols

A	Area(m^2)
C	Constant
C_m	Constant (m = 1, 2, 3…)
C_f	Fraction
C_p	Specific heat($J \cdot kg^{-1} \cdot K^{-1}$)
C_{pd}	Pressure drops characteristic coefficient
D_h	Hydraulic diameter(m)
f	Friction factor or resistance coefficient
g	Gravitational acceleration($m \cdot s^{-2}$)
G	Mass flow density($kg \cdot m^{-2} s^{-1}$)
Gr	Grashof number
h	Specific enthalpy($J \cdot kg^{-1}$)
m	Mass flow rate (kg/s)
Nu	Nusselt number
P	Pressure (Pa)
$P_{h,}$	Heated perimeter(m)
Pr	Prandtl number
q_g	Heat generation($W \cdot m^{-3}$)
q_{hf}	Heat flux($W \cdot m^{-2}$)
q_l	Linear power($W \cdot m^{-1}$)
Re	Reynold number
t	Time(s)
T	Temperature(K)
U	Effective momentum velocity($m \cdot s^{-1}$)
x	Co-ordinates(m)
z	Co-ordinates(m)
ε	Iterative convergence constant
ϕ^2	Two-phase friction multiplier
λ	Heat conductivity($W \cdot m^{-1} \cdot K^{-1}$)
μ	Dynamic viscosity($kg \cdot m^{-1} \cdot s$)
θ	Angle(rad)
‘	Density($kg \cdot m^{-3}$)
δ	Thickness(m)

Subscripts

av	Average
c	Cladding
f	Fuel
in	Inlet
int	Internal
m	Horizontal node number
n	Axial node number
u	Fuel pellet
w	Wall
0	Initial

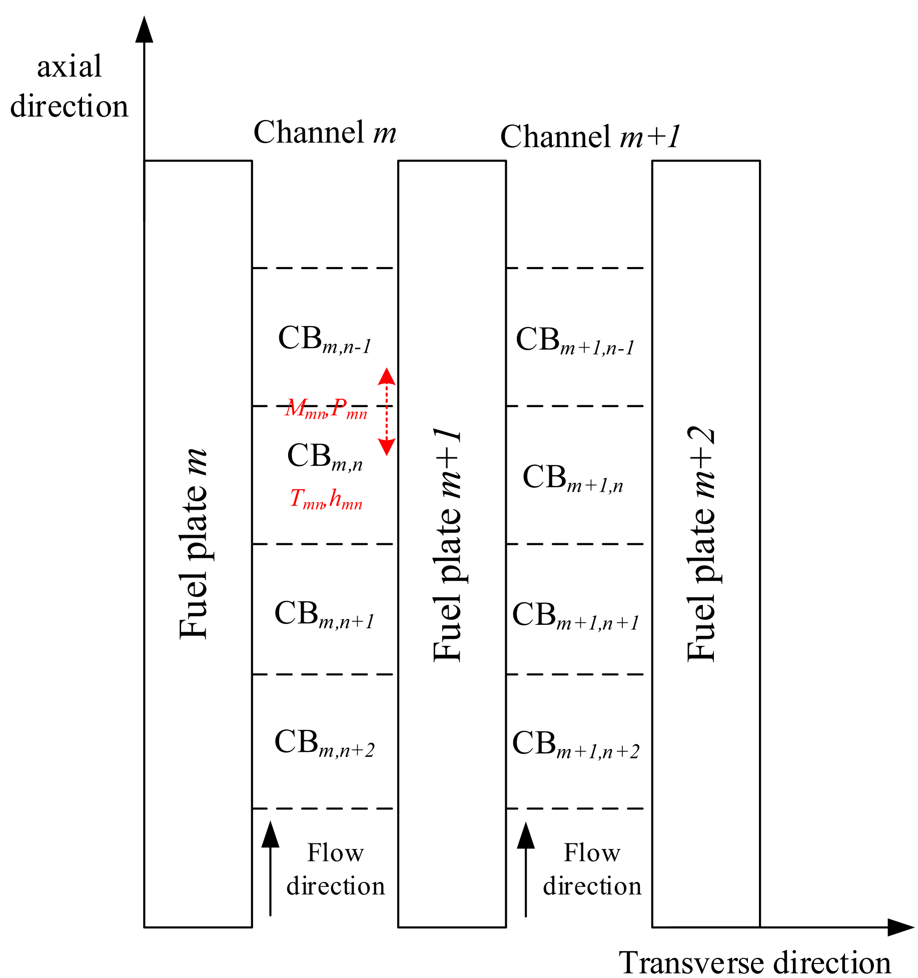

Fig. 1. The division of the domains of the fuel plate and the flow channel (cross section).

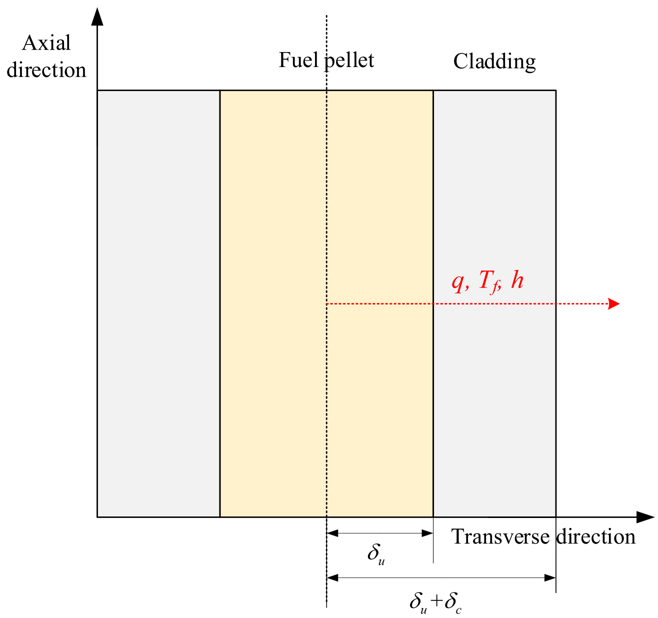

Fig. 2. Schematic diagram of plate fuel element (cross section).

reactor under accident conditions (such as positive reactivity insertion). Forti (1970) developed the COSTAX-BOIL code for axial dynamics analysis of reactor transients. This code was originally used to analyze traditional boiling water reactors and pressurized water reactors and was later successfully used to analyze the flow blocking accident of IAEA 10 MW MTR.

3. Thermal hydraulic system code. RELAP5 is the most widely used thermal–hydraulic system code developed by the Idaho National Engineering Laboratory. Many scholars have used different versions of RELAP5 to perform steady-state and transient analysis on research reactors, such as RELAP5/MOD3.2 (Hamidouche et al., 2004), RELAP5/MOD3.3 (Lu et al., 2009) and RELAP5/MOD3.4 (Guo et al., 2018). In addition, based on another system code ATHLET, Koppers (Koppers and Koch, xxxx) proposed a new modeling method for plate fuel elements.
4. The subchannel analysis code. COBRA-3C can be used to calculate plate-type fuel elements, but the models and correlations in the code are outdated (Chao et al., 1980). Based on the homogenous equilibrium model and the balanced drop pressure approach, Castellanos-Gonzalez et al. (Castellanos-Gonzalez et al., 2018) developed the subchannel code COTENP for plate-type fuel elements and validated it with CARR experimental data. Almachi (Almachi et al., 2022) innovatively coupled the subchannel analysis code Subchanflow and the neutron analysis code Monte Carlo Serpent 2, achieving for the first time transient and steady-state analysis of the IAEA 10 MW benchmark at the subchannel and plate levels.

Table 1
Correlations used in SACOS-PLATE: heat transfer correlations.

Species	Correlation	References
Single phase flow heat transfer	Dittus-Boelter $Nu = 0.023\mathrm{Re}_f^{0.8}Pr_f^{0.4}$	(Dittus & Boelter, 1985)
	Colburn $Nu = 0.023\mathrm{Re}_f^{0.8}Pr_f^{1/3}$	(Spurgeon, 1969)
	Sieder-Tate $Nu = 0.027\mathrm{Re}_f^{0.8}Pr_f^{1/3}\left(\frac{\mu_f}{\mu_w}\right)^{0.14}$	(Sieder & Tate, 1936)
	Mikhayev $Nu = 0.021\mathrm{Re}_f^{0.8}Pr_f^{0.43}\left(\frac{Pr_f}{Pr_w}\right)^{0.25}$	(Wallis, 2020)
	Collier $Nu = 0.17\mathrm{Re}_f^{0.33}Pr_f^{0.43}\left(\frac{Pr_f}{Pr_w}\right)^{0.25}Gr^{0.1}$Sudo $Nu = \frac{0.0296Re^{0.8}Pr}{\left[1+1.54Pr^{-\frac{1}{4}}Re^{-0.1}(Pr-1)\right]}$	(Wallis, 2020) (Sudo et al., 1990)
	Gnielinski $Nu = \frac{\left(\frac{f}{8}\right)(\mathrm{Re}_f-1000)Pr_f}{1+12.7\left(\frac{f}{8}\right)^{0.5}\left(Pr_f^{2/3}-1\right)}$	(Gnielinski, 1975)(Gnielinski, 1975)
	Petukhov $Nu = \frac{\left(\frac{f}{8}\right)\mathrm{Re}_f Pr_f}{1.07+12.7\left(\frac{f}{8}\right)^{0.5}\left(Pr_f^{2/3}-1\right)}\left(\frac{\mu_f}{\mu_w}\right)^{0.11}$	(Petukhov, 1970)
Subcooled and saturated boiling heat transfer	Chen $h = 0.023F\left[\frac{G(1.0-x)De}{\mu_f}\right]^{0.8}\left[\frac{\mu Cp}{k}\right]_f^{0.4}\left(\frac{k_f}{De}\right)+$ $0.00122S\left[\frac{k_f^{0.79}C_{pf}^{0.45}\rho_f^{0.49}}{\sigma^{0.5}\mu_f^{0.29}h_{f\hat{B}}^{0.24}\rho_g^{0.24}}\right](T_w-T_S)^{0.24}(p_w-p_s)^{0.75}$	(Wallis, 2020)
Film boiling heat transfer	Polimik $Nu = \mathrm{a}\left\{\mathrm{Re}_g\left[x+\frac{\rho_g}{\rho_f}(1-x)\right]\right\}^{0.9}$	(Wallis, 2020)
	Mattson $h = \mathrm{aRe}_g^{0.244}(\mathrm{Pr}_g)_w^{2.54}D_e^{-0.304}\lambda_g^{0.334}$	(Wallis, 2020)

Table 2
Correlations used in SACOS-PLATE: other correlations.

Species	Correlation	References
Single-phase friction resistance coefficient	McAdams $f = \frac{0.184}{\mathrm{Re}^{0.2}}$	(Rust, 1979)
	Blasius $f = \frac{0.3164}{Re^{0.25}}$	(Kandlikar et al., 2005)
	Troniewski-Ulbrich $f = \frac{64}{\mathrm{Re}(2\alpha)^{0.16}}$	(Troniewski & Ulbrich, 1984)
	Nikuradze $\frac{1}{f^{0.5}} = -0.4 + 4.0\log\left(\mathrm{Re}\cdot f^{0.5}\right)$	(Idelchik, 1986)
	Cole-Brook $\frac{1}{\sqrt{f}} = -2\log\left(\frac{\varepsilon/D_e}{3.71} + \frac{2.51}{\mathrm{Re}\sqrt{f}}\right)$	(Colebrook et al., 1939)
Two-phase friction multiplier	Martinelli-Nelson $\Phi_{fo}^2 = 1.0 + 1.2x^{0.75\left(1.0+0.01\sqrt{\rho_f/\rho_g}\right)}\left[\left(\frac{\rho_f}{\rho_g}\right)^{0.8} - 1\right]$	(Sher, 1956)
	Friedel-Zhang $\Phi_{lo}^2 = (1-x)^2 + 2.87x^2\frac{p_{cr}}{p} + 1.68x^{0.8}(1-x)^{0.25}\left(\frac{p_{cr}}{p}\right)^{1.64}$	(Friedel, 1979; Zhang and Webb, 2001)
	Chisholm-Tran $\Phi_{lo}^2 = 1 + (4.3\Gamma^2 - 1)\left[N_{Conf}x^{0.875}(1-x)^{0.875} + x^{1.75}\right]$	(Chisholm, 1983; Tran et al., 2000)
Critical heat flux	Mirshak $q = 151(1 + 0.1198v)(1 + 0.00914\Delta T_{sub})(1 + 0.19p)$	(International, 1980; Kaminaga, 1990)
	Labunstov $q_{CHF} = 145.4G\left[1 + 2.5V^2/G\right]^{1/4}\left(1 + \frac{15.1C_p\Delta T_{sub}}{\lambda P^{1/2}}\right)$	(International, 1980)
	Sudo $q_{CHF1}^* = 0.005\lvert G^*\rvert^{0.611}$ $q_{CHF2}^* = (A/A_H)\Delta T_{\mathrm{subin}}^* \lvert G^*\rvert$ $q_{CHF3}^* = 0.7\frac{A}{A_H}\frac{(W/\lambda)^{0.5}}{\left[1 + (\rho_g/\rho_l)^{0.25}\right]^2}$ $q_{CHF4}^* = 0.005\lvert G^*\rvert^{0.611}\left(1 + \frac{5000}{\lvert G^*\rvert}\Delta T_{sub,0}^*\right)$ $G_1^* = \left(\frac{0.005}{(A/A_H)\Delta T_{sub,in}^*}\right)^{2.5707}$	(Sudo et al., 1985)

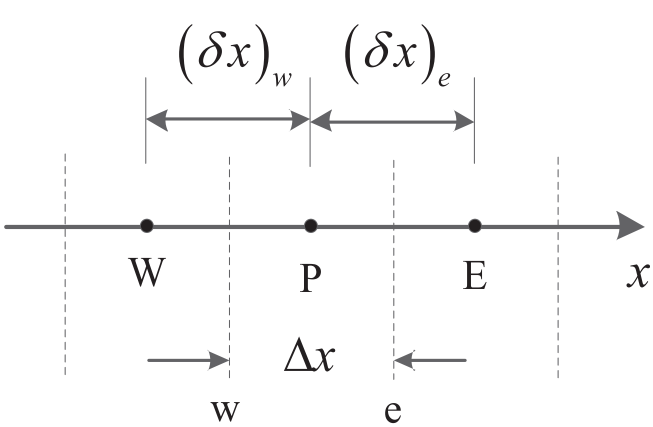

Fig. 3. Plate fuel node grid relationship.

The Nuclear Reactor Thermal-hydraulic Laboratory of Xi'an Jiaotong University (XJTU-NuTheL) has been committed to the thermal–hydraulic numerical simulation of nuclear reactor cores and the development of related professional codes for many years. In recent years, a sequence of *Subchannel Analysis Codes Of Safety* (SACOS) have been developed for different types of reactors (Gui et al., 2021a). Chaudri et al. (Chaudri et al., 2012; Chaudri et al., 2013) proposed the subchannel code SACOS-SCWR for supercritical water reactor (SCWR) and it has been applied to high performance light water reactor (HPLWR) neutron/thermal coupled analysis. While Wu et al. (Wu et al., 2013) performed thermal–hydraulic subchannel analysis with modified constitutive model on the sodium-cooled fast reactor (SFR) based on the SACOS-SFR code. Taking into account the influence of inter-wrapper flow (IWF) on the nuclear core, Yue et al. (2018) added and validated a subsequent two-dimensional layered IWF model. Wang et al. (2013) extended the flow and heat transfer model of liquid lead–bismuth as well as the calculation and analysis capabilities of thermodynamic characteristics of advanced lead–bismuth fast reactor (ALBFR) on the basis of the original SACOS. Afterwards, several auxiliary models in the specification were further optimized by Wang et al. (2018) and Cai et al. (2016) taken the effects of the additional source term on the thermal–hydraulic phenomena into account during nuclear reactor core operation. Gui et al. (2021a) optimized the governing equations and algorithms of the original subchannel analysis code, which significantly improved the computational convergence and accuracy of the code. According to the special thermal–hydraulic phenomena of lead-based fast reactor (LFR), Wu et al. (2021) compared and evaluated the flow and heat transfer correlations based on the collected experimental data, and then developed the subchannel analysis code SACOS-PB for LFR. Recently, Gui et al. (2021b) developed the input preprocessing and parallel capabilities of the SACOS code and validated the greatly improved computational efficiency through calculation examples.

In this paper, the subchannel analysis code SACOS-PLATE of the plate fuel element is developed and validated by comparing with the data in relevant literatures. In the second chapter, the background and physical model of SACOS-PLATE are introduced. In the third chapter, the experimental data of the MTR research reactor is used to validate the flow and heat transfer model of the developed subchannel code. In the fourth chapter, based on the design parameters of the JRR-3 research reactor, the whole component is simulated and compared with other subchannel codes. In the fifth chapter, the experimental data from the Bettis test is used to validate the code. In the sixth chapter, the preliminary research results of this research are summarized, and the future research directions are prospected.

2. Code description

The SACOS-PLATE subchannel analysis code is one of the updated versions based on the subchannel analysis code SACOS (Gui et al., 2021a,b), which is applicable for rod bundle fuel elements, by modifying the physical model. As the governing equations and constitutive model of SACOS were thoroughly described in our previous work, they will not be redundantly elaborated in this paper. Interested readers are encouraged to refer to reference (Gui et al., 2021a,b) for further details. The main modified models in SACOS-PLATE are:

1. The heat conduction model of the fuel plate. Unlike the bundle fuel elements, the heat distribution on both sides of the fuel plate is generally unequal because the maximum temperature point of the plate fuel element is often not at the center of its geometry. Therefore, it is necessary to analyze and calculate the hottest spot and obtain the two-dimensional temperature field distribution inside the fuel plate.
2. Flow distribution model. For rectangular flow channels, there is generally no lateral mixing in adjacent channels, so the momentum

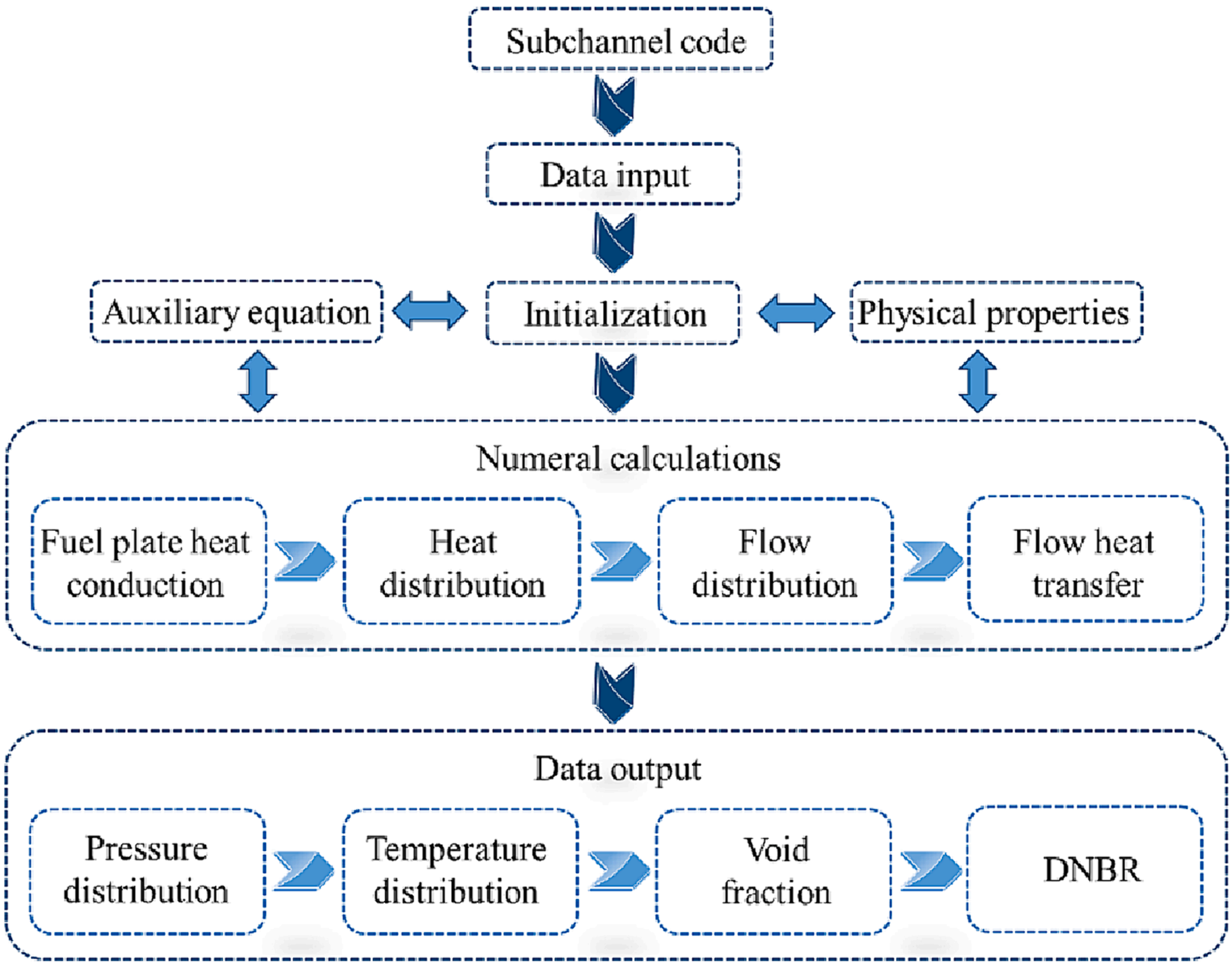

Fig. 4. Code structure diagram.

equation and the coupling relationship between flow and pressure drop need to be modified.

3. Flow heat transfer model. Although many widely used correlations are applicable to both rod bundle and plate-type fuel elements, SACOS-PLATE still adds a large number of flow and heat transfer correlations for plate-type fuel elements.

2.1. Subchannel and control volume division

For plate-type fuel elements, the internal structure of the core is compact, restricting the flow of coolant. Numerous fuel elements divide the core coolant flow area into many parallel coolant channels. Although it has been achieved to divide all parallel channels in the core into subchannels for analysis (Almachi et al., 2022), it is not an easy task. Commonly used subchannel division methods are as follows:

1. According to the geometry and power distribution symmetry of the entire core, select several subchannels for calculation
2. Multi-stage division: First divide the entire core by fuel assembly to obtain the fuel assembly with the highest temperature; then divide the inner subchannels of the hottest fuel assembly to obtain the hottest subchannels.
3. Combination division: Empirically and theoretically, subchannels near where the hottest fuel assemblies are likely to occur are finely divided, while subchannels far away from them are roughly divided.
4. Directly perform plate/subchannel level simulations of the entire core.

The four subchannel division methods also have their own pros and cons. The first method has certain limitations and the number of subchannels divided is large. The third method relies on personal experience and requires the entire core to be involved. The second method is a calculation method from coarse to fine, which requires two analysis calculations, and the process is more complicated. The fourth method requires modeling of the entire reactor core, which increases the difficulty of modeling and the amount of calculation. However, with the rapid development of software and hardware in the past decade, full-core subchannel level simulations will become mainstream. At the same time, this approach will be more used to couple with other codes such as neutron analysis codes (Almachi et al., 2022), making it possible to refine numerical reactors. Our team is also currently developing a subchannel analysis code based on the MOOSE framework. For the subchannel division of plate-type fuel elements in this study, method two can be used. First, perform rough calculations on the entire core, and then perform detailed subchannel analysis and calculation on the specified fuel assembly (generally the hottest fuel assembly).

The first step in the numerical calculation of the flow and heat transfer problem is the area discretization, that is, the continuous calculation area in space is divided into several domains. The second step is the discretization of the governing equations, that is, the partial differential equations describing the flow and heat transfer problems are transformed into a set of algebraic equations. The subchannel analysis

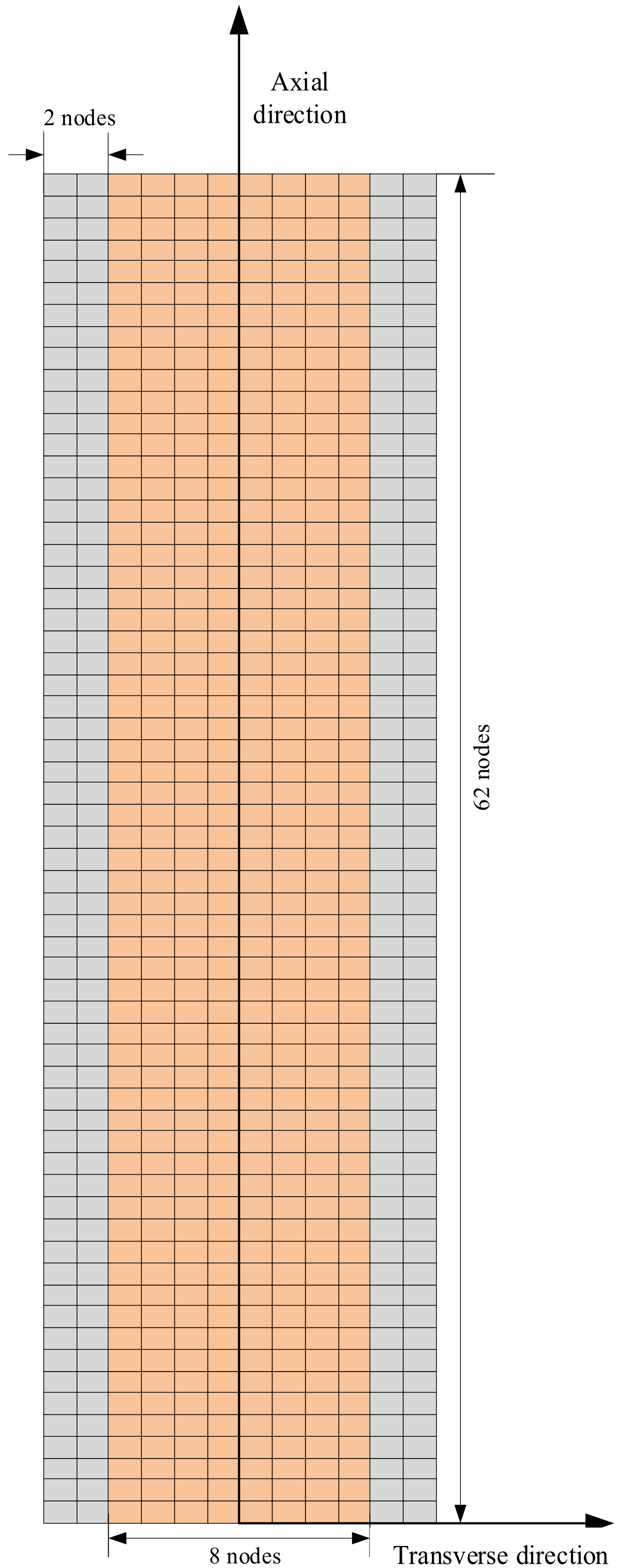

Fig. 5. Schematic diagram of the control volumes division of the experimental section.

Table 3
System parameters under different working conditions in RA-6 experiment (Almachi et al., xxxx; Sillin et al., 2009).

System parameter	Experimental conditions 1	Experimental conditions 2
Power(*kW*)	19	25
Coolant mass flow rate(*kg/s*)	0.12	0.16
Inlet pressure (*MPa*)	0.17	0.17
Inlet temperature (*oC*)	38	38
Reynolds range	$7 \times 10^3 < Re < 1.4 \times 10^4$	$7 \times 10^3 < Re < 1.4 \times 10^4$
Cladding thermal conductivity ($W{\cdot}m^{-1}{\cdot}K^{-1}$)	180	180
Cladding specific heat ($J{\cdot}kg^{-1}{\cdot}K^{-1}$)	892	892
Cladding density($kg{\cdot}m^{-3}$)	2700	2700
Fuel thermal conductivity ($W{\cdot}m^{-1}{\cdot}K^{-1}$)	15	15
Fuel specific heat($J{\cdot}kg^{-1}{\cdot}K^{-1}$)	130	130
Fuel density($kg{\cdot}m^{-3}$)	8400	8400

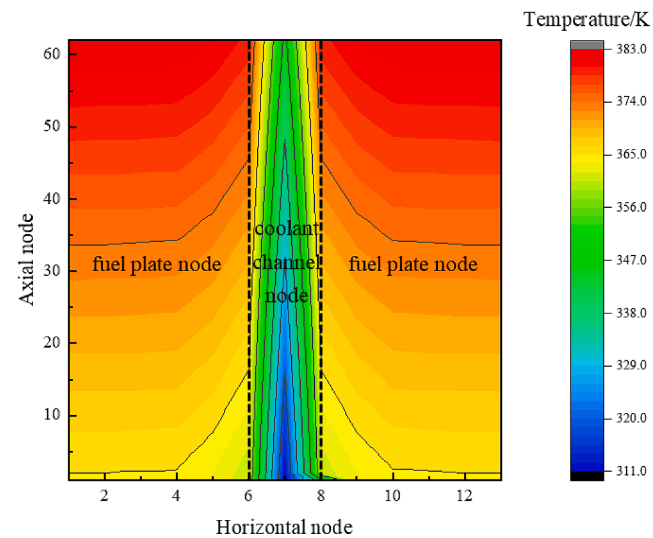

Fig. 6. Two-dimensional temperature field distribution of fuel plate (cross section).

computational domain adopts a staggered grid, as illustrated in Fig. 1. The axial flow rate and pressure are defined on the boundary of the domain, and the state parameters such as specific enthalpy, density and cavitation share are defined in the center of domain. Additionally, a second-order upwind scheme is employed for the convection term, and the relaxation factor is set to 0.7 by default.

2.2. Basic governing equations

Based on the aforementioned division of subchannels and control volumes, the basic governing equations of SACOS-PLATE can be obtained as follows.

2.2.1. Mass conservation equation

The mass conservation equation of the mixture in the control volume (m, n) is as follows:

$$A_m \frac{\Delta z_n}{\Delta t}\left(\rho_{mn} - \rho'_{mn}\right) + m_{mn} - m_{mn-1} = 0 \tag{2.1}$$

Where A is axial flow area, ρ is density, m is mass flowrate. The superscript of ρ'_{mn} indicates that this is the physical quantity of the last time step.

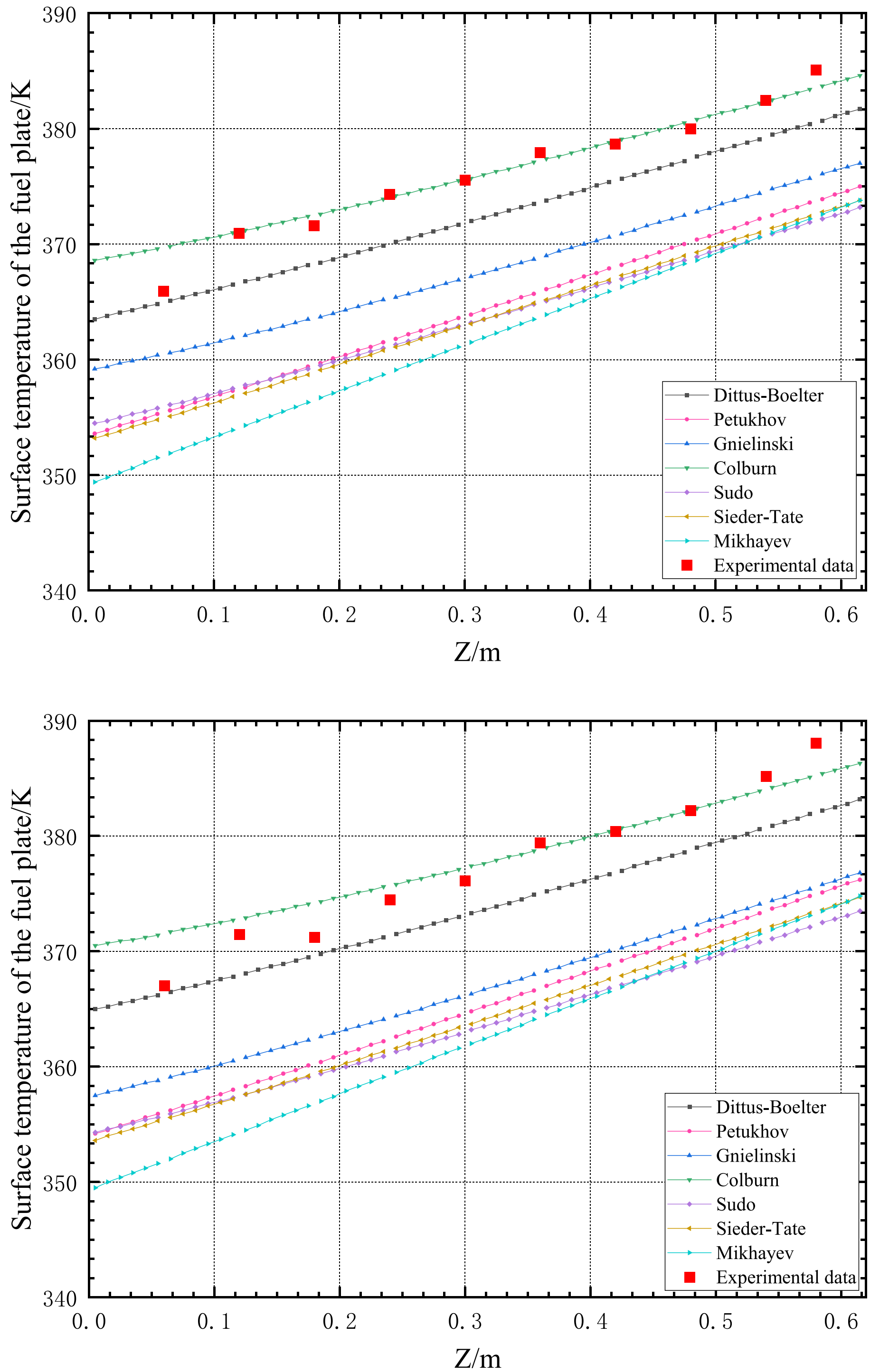

Fig. 7. Comparison of simulation results and experimental data of fuel plate surface temperature (a) Power is 19 kw; (b) Power is 25 kw (Almachi et al., 2021).

Table 4
Root-mean-square deviations of different single-phase heat transfer Correlations.

Experimental conditions	Dittus-Boelter	Petukhov	Gnielinski	Colburn	Sudo	Sieder-Tate	Mikhayev
1	3.78 K	12.13 K	8.66 K	1.43 K	12.91 K	12.99 K	14.32 K
2	3.84 K	12.53 K	10.90 K	2.20 K	14.28 K	13.69 K	16.43 K

2.2.2. Momentum equation

The momentum equation of the mixture in the control volume (m, n) is as follows:

Table 5
JRR-3 standard fuel assembly dimensions (Albati et al., 2014).

Type of fuel assembly	Standard fuel assembly
No. of fuel assemblies	26
No. of fuel plates per fuel element	21
Plate length (*mm*)	770
Plate width (*mm*)	62
Plate thickness (*mm*)	0.51
Cladding thickness (*mm*)	0.38
Channel thickness (*mm*)	2.35
Channel flow width (*mm*)	66.6

Table 6
Design input parameters of JRR-3. (Albati et al., 2014).

Type of parameters	Forced convection
Total Power (*MW*)	20
Total heated area of the core (m^2)	55.6
Total core flow (*kg/s*)	662.67
Coolant inlet velocity (*m/s*)	5.83
Core inlet temperature (°C)	35
Core inlet pressure (*MPa*)	0.155

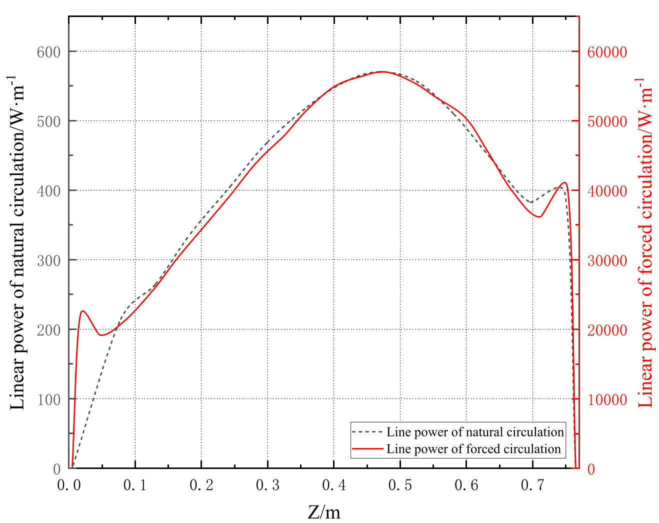

Fig. 8. The axial line power distributions of the nuclear reactor (Albati et al., 2014).

$$\frac{\Delta z_n}{\Delta t}\left(m_{mn}-m_{mn}^{'}\right)+m_{mn}U_{mn}-m_{mn-1}U_{mn}=$$
$$-A_m(P_{mn}-P_{mn-1})-gA_m\Delta z_n\rho_{mn}\cos\theta-\frac{1}{2}\frac{\Delta z_n f\phi^2}{D_h\rho_{mn}}|m_{mn}|\frac{m_{mn}}{A_m} \tag{2.2}$$

Where U is effective momentum velocity, P is pressure, g is gravity acceleration, θ is the inclination angle of the channel, f is friction factor, ϕ^2 is two-phase friction multiplier, D_h is hydraulic diameter.

2.2.3. Energy conservation equation

The energy conservation equation of the mixture in the control volume (*m, n*) is as follows:

$$\frac{A_m}{\Delta t}\left[\rho_{mn}\left(h_{mn}-h_{mn}^{'}\right)+h_{mn}\left(\rho_{mn}-\rho_{mn}^{'}\right)\right]$$
$$+\frac{1}{\Delta z_n}\left(m_{mn}h_{mn}-m_{mn-1}h_{mn-1}\right)=\sum_{k=left,right}\left(P_{h,km}q_{hf,km}+C_{f,km}q_{l,km}\right) \tag{2.3}$$

Where h is the specific enthalpy, $P_{h,km}$ is the heated perimeter between channel m and fuel plate k ($k = left, right$), q_{hf} is the heat flux from a fuel plate into the fluid, C_f is the fraction of the fission power that enters the coolant directly, q_l is linear power.

2.3. Constitutive equations

2.3.1. Heat conduction model of plate fuel element

In a nuclear reactor, the energy generated by fission is transferred to the coolant through the cladding, and then the coolant flows out of the core through forced circulation. Fig. 2 shows a schematic diagram of the structure of a plate fuel element, in which the thickness of the fuel pellet and the cladding are δ_u and δ_c, respectively. In order to establish the plate fuel heat conduction equation, the following hypotheses need to be proposed:

1. The plate fuel element is centrally symmetrical.
2. The internal heat source of the cladding material is neglected.
3. Ignore the gap between the fuel pellet and the cladding, that is, the cladding and the fuel pellet are in close contact.
4. Since the plate fuel element is wide and long, the axial heat conduction of the core is ignored, and only the lateral heat conduction is considered.

The heat conduction model of the plate-type fuel element is a one-dimensional heat conduction model, and its heat conduction equation is:

$$(\rho c_p)_u\frac{\partial T_u}{\partial t}=\frac{\partial}{\partial x}\left(\lambda_u\frac{\partial T_u}{\partial x}\right)+q_g \tag{2.4}$$

$$(\rho c_p)_c\frac{\partial T_c}{\partial t}=\frac{\partial}{\partial x}\left(\lambda_c\frac{\partial T_c}{\partial x}\right) \tag{2.5}$$

The gap between the pellet and the cladding is neglected, the boundary conditions between the inner surface of the cladding and the fuel pellet are:

$$\begin{cases} T_u|_{x=\delta_u}=T_c|_{x=\delta_u} \\ \lambda_u\frac{\partial T_u}{\partial x}|_{x=\delta_u}=\lambda_c\frac{\partial T_c}{\partial x}|_{x=\delta_u}\end{cases} \tag{2.6}$$

The boundary conditions between the outer surface of the cladding and the coolant are:

$$-\lambda_c\frac{\partial T_c}{\partial x}|_{x=\delta_c+\delta_u}=h\left(T_c|_{x=\delta_c+\delta_u}-T_f\right) \tag{2.7}$$

It is worth mentioning that the physical properties used in the above equations are functions of temperature.

2.3.2. Flow distribution model

The flow distribution between the components is determined by the total pressure drop of each component channel. Due to the operating conditions of the core, the pressure in the lower and upper chambers is constant, that is, the total pressure drop of each component channel is the same:

$$\Delta P_m=\Delta P_{m+1}=\cdots=Constant \tag{2.8}$$

Where ΔP_m is the total pressure drop of component channel m.

Due to the large geometry of the core, the pressure distribution at the inlet of each fuel assembly is not uniform. In order to characterize the unevenness of the inlet pressure of the channel, the channel inlet resistance coefficient $f_{m,in}$ (for channel m) is introduced, which is generally given by experimental measurement or empirical data. Therefore, the total pressure drop of the channel can be divided into inlet pressure drop and internal pressure drop:

$$\Delta P_m=\Delta P_{m,in}+\Delta P_{m,int} \tag{2.9}$$

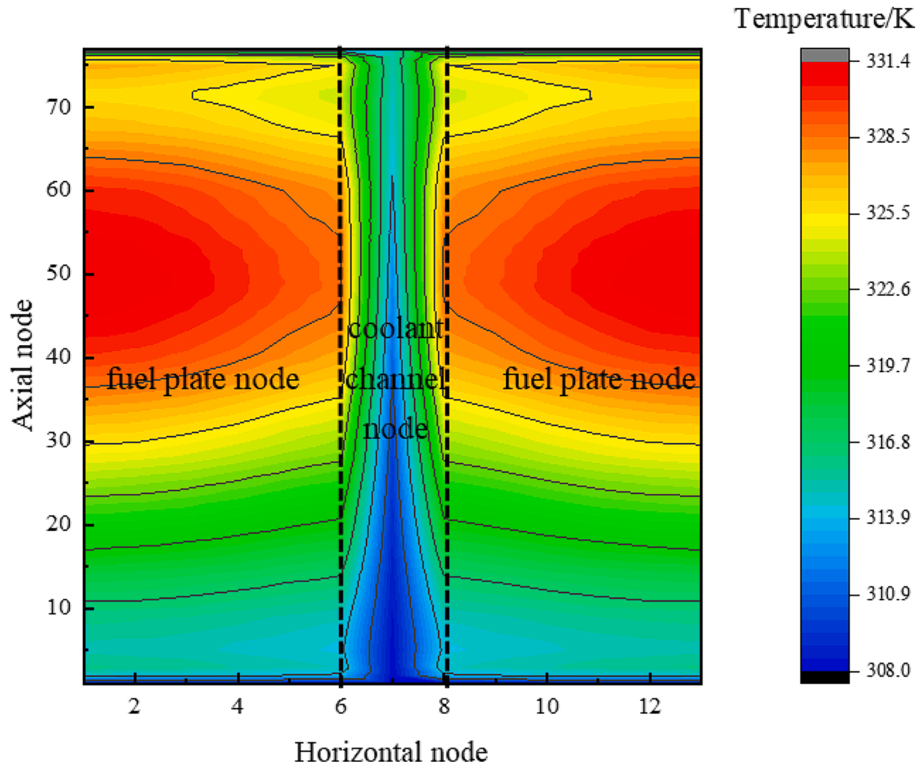

Fig. 9. Two-dimensional temperature field distribution of fuel plate (cross section).

$$\Delta P_{m,in} = f_{m,in} m_m^2 \tag{2.10}$$

$$\Delta P_{m,int} = \sum_Z \left[\rho_{mn} g \Delta z_n + \frac{1}{2}\frac{f m_m^2}{A_m^2 D_h \rho_{mn}} \Delta z_n + \frac{m_m^2}{A_m^2}\Delta\left(\frac{1}{\rho_{mn}}\right)\right] \tag{2.11}$$

In the calculation of flow distribution, the initial distribution of flow is given according to the channel area:

$$m_{m,0} = G_{m,0} A_m \tag{2.12}$$

Where $G_{m,0}$ is the initial mass flow density (kg m^{-2} s^{-1}). The assumed initial flow distribution is not the actual flow distribution, so it is necessary to modify the assumed flow distribution according to the calculation results. The total pressure drops of each channel ΔP_i was calculated by solving the thermo-hydraulic system of equations. Then the average pressure drops of all channels ΔP_{av} is calculated by weighting the channel area:

$$\Delta P_{av} = \frac{\sum_M A_m \Delta P_m}{\sum_M A_m} \tag{2.13}$$

The total pressure drops of each channel ΔP_i is compared with the average pressure drop of all flow channels ΔP_{av}. If the pressure drop does not meet the convergence condition, the flow rate of each channel needs to be adjusted, that is, the core flow rate is redistributed, and then the calculations are performed again. Finally, the total pressure drop of the component channels is calculated iteratively until the coolant flow pressure drop in all the channels meets the convergence condition.

$$\frac{|\Delta P_m - \Delta P_{av}|}{\Delta P_{av}} < \varepsilon \tag{2.14}$$

Where ε is the iterative convergence constant ($\varepsilon = 0.2\backslash\%$ in the code).

From formulas (2.9) to (2.11), it can be deduced that the total pressure drop has the following relationship with the inlet flow rate:

$$\Delta P_{m,int} - \overline{\rho}_m g z = \left\{ f_{m,in} + \sum_Z \left[\frac{1}{2}\frac{f}{A_m^2 D_h \rho_{mn}} \Delta z_n + \frac{1}{A_m^2}\Delta\left(\frac{1}{\rho_{mn}}\right)\right]\right\} m_m^2 \tag{2.15}$$

In order to simplify the above formula, the pressure drop characteristic coefficient $C_{pd,m}$ (The unit of $C_{pd,m}$ is the same as $f_{m,in}$, which is kg^{-1}·m^{-1}) is introduced:

$$C_{pd,m} = f_{m,in} + \sum_Z \left[\frac{1}{2}\frac{f}{A_m^2 D_h \rho_{mn}} \Delta z_n + \frac{1}{A_m^2}\Delta\left(\frac{1}{\rho_{mn}}\right)\right] \tag{2.16}$$

Therefore, the flow rate of the previous time step m' and the flow rate of the current time step after the pressure correction $\widehat{m}$ are shown as follows:

$$C_{pd,m} m_m'^2 = \Delta P_{m,int} - \overline{\rho}_m g z \tag{2.17}$$

$$C_{pd,m} \widehat{m}_m^2 = \Delta P_{av} - \overline{\rho}_m g z \tag{2.18}$$

Finally, the flow rate of this time step is obtained by correcting the total inlet flow rate:

$$m_m = \widehat{m}_m \frac{m_{total}}{\sum_M \widehat{m}_m} \tag{2.19}$$

2.3.3. Flow heat transfer model

To obtain the complete thermal–hydraulic characteristics of the core/component subchannels, it is necessary to supplement a series of auxiliary models to achieve the mathematical closure of the governing equations in addition to the basic mathematical and physical models. The auxiliary models mainly include heat transfer coefficient model, flow resistance coefficient model and critical heat flux model. Compared with rod fuel elements, the thermal–hydraulic characteristics of the rectangular narrow channels formed by the plate fuel elements are different, so a large number of correlations applicable to the plate fuel elements are added to SACOS-PLATE, as shown in Table 1 and Table 2. It is worth noting that the applicable conditions of all correlations are different, and the applicable scope of each correlation will not be described in detail in this article (please see references listed in Table 1 and Table 2 for details).

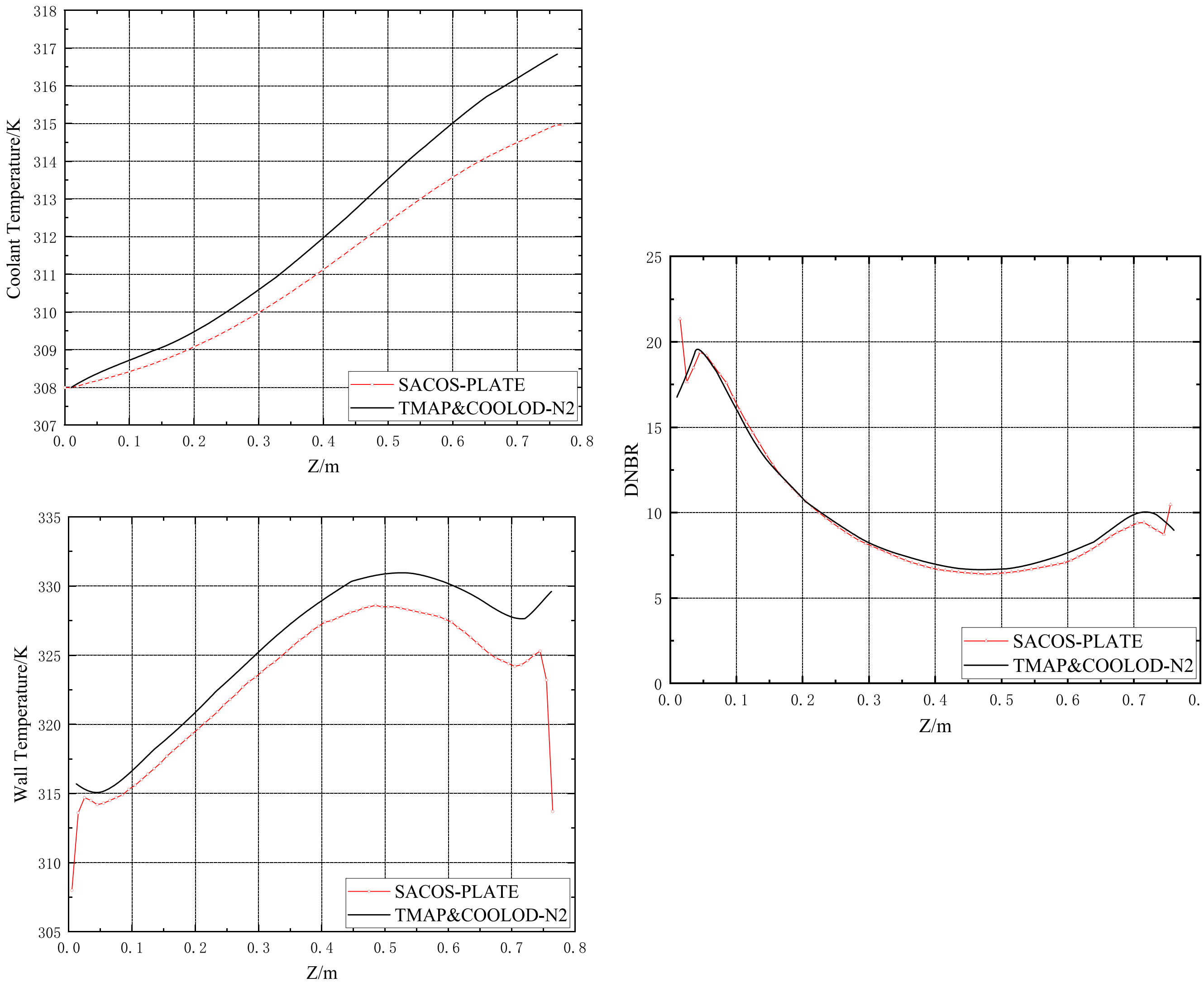

Fig. 10. Forced convection cooling analysis results (a) bulk temperature; (b) wall temperature; (c) DNBR (Albati et al., 2014).

Table 7
Design input parameters of Bettis test (Jacket et al., 1958).

Parameters	Values
Flow channel geometry (*mm*)	1.27 × 25.4 × 306.4
Inlet pressure (*MPa*)	13.8
Fuel Thermal Conductivity (*W/m/k*)	32
Cladding thermal conductivity (*W/m/k*)	170

Table 8
Parameters of the cases.

Number	Inlet Subcooling (K)	Mass velocity (kg/m^2/s)	CHF (MW/m^2)
53	166.11	1246.39	2.41
54	157.22	1830.94	3.53
55	128.89	215.64	0.915
56	130.56	462.48	1.35
57	129.44	575.05	1.59
58	127.78	901.91	1.86
59	129.44	1076.86	2.17
60	127.22	1529.85	2.61
61	130.00	1790.25	2.90
52	187.22	1993.69	4.48

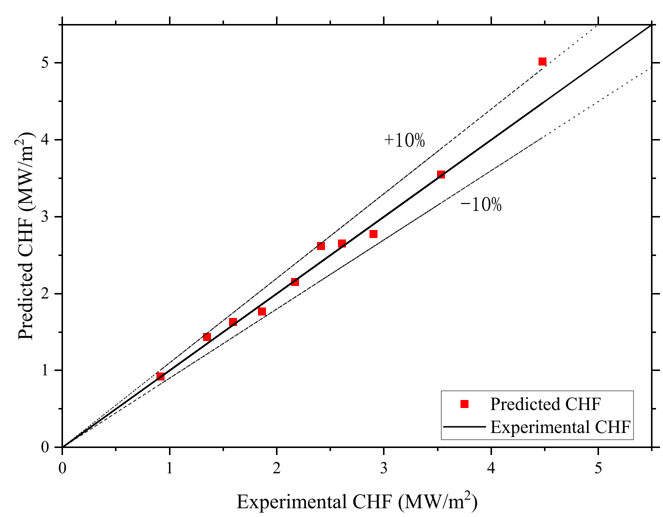

Fig. 11. Comparison curve between the predicted results of SACOS-PLATE and the experimental results.

2.4. Discretization of the heat conduction equation

The discrete form of the thermal conductivity equation of plate fuel is derived by using the control volume integral method, and the integral of the control volume P (as shown in Fig. 3) in the time interval $[t, t+\Delta t]$ can be obtained:

$$\int_t^{t+\Delta t}\int_w^e \rho c\frac{\partial T}{\partial t}dxdt = \int_t^{t+\Delta t}\int_w^e \frac{\partial}{\partial x}\left(\lambda\frac{\partial T}{\partial x}\right)dxdt + \int_t^{t+\Delta t}\int_w^e q_g dxdt \tag{2.20}$$

The completely implicit discrete equation of heat conduction of fuel plate is derived:

$$a_P T_P = a_E T_E + a_W T_W + T_P^0 a_P^0 + b \tag{2.21}$$

where $a_E = \frac{\lambda_e}{(\delta x)_e}$, $a_W = \frac{\lambda_w}{(\delta x)_w}$, $a_P^0 = \frac{(\rho c)_P \Delta x_P}{\Delta t}$, $a_P = a_E + a_W + a_P^0$, $b = q_g \Delta x_P$。 **

SACOS-PLATE mainly consists of six structural modules: data input, initialization, auxiliary equations, physical property parameters, numerical calculation and data output, as shown in Fig. 4.

Pressure gradient algorithm is adopted in SACOS-PLATE. First coolant is calculated from the energy equation of enthalpy, axial pressure gradient is calculated by the momentum equation, at the same time, through continuous equation to calculate the axial mass flow rate, the last known enthalpy and steam content under the condition of correlation through the steam content-void fraction, and calculate the void fraction of coolant density, complete the calculation of an iteration step.

3. Validation of code using RA-6 test data

3.1. Brief introduction of RA-6 experiment

The RA6 experiment was a 1:1 simulation experiment for the MTR research reactor conducted in Argentina from 2000 to 2005, which studied the thermal behavior of MTR fuel assemblies under transient conditions and provided experimental data for code validation and improvement (García et al., 2008).

In RA-6 experiment, the subchannel consists of a rectangular section with 2.7 mm thick and 60 mm wide. The height of the experimental device is 76 cm, of which 62 cm is the heating section. The spacing between the aluminum plates is 2.7 mm, and thermal insulation and electrical insulation are adopted. The thickness of the heating aluminum plate is 5 mm or 6 mm (García et al., 2008; Sillin et al., 2009).

3.2. SACOS-PLATE geometrical model of RA-6 test section

In the modeling, each coolant channel is axially divided into 62 nodes (test section 620 mm). The fuel plate is divided into 12 nodes (8 fuels and 2 cladding) in the transverse direction, as shown in Fig. 5. In addition, other boundary conditions (pressure, mass flow and coolant temperature) are directly used from RA-6 experiment, as shown in Table 3.

3.3. Simulation and analysis

Seven different heat transfer equations including Dittus-Boelter, Petukhov, Gnielinski, Colburn, Sudo, Sieder-Tate and Mikhayev were used to carry out seven simulation calculations. In these 7 calculations, other constitutive models are consistent. The single-phase friction resistance coefficient correlation is Blasius correlation and the CHF correlation is the Sudo correlation. Both the experiment and the code simulation are in a steady state. Since the experiment is heated by electric wires, the power is evenly distributed.

In order to validate the calculation results of the SACOS-PLATE code, the data of the RA-6 tests are compared with the simulation results of the code. Fig. 6 shows the two-dimensional temperature field distribution of the fuel plate simulated by the code. In the horizontal direction, it can be seen that the temperature on both sides of the left and right fuel plates is the highest, and the center temperature is the lowest. This is because there is a heat insulating material on the outside of the fuel plate, which prevents the temperature diffusion. In the middle of the two fuel plates is the coolant channel, so the temperature is the lowest. In the axial direction, the temperature of the fuel plate continues to rise along the direction of coolant flow, and the rise is basically the same everywhere. This is because the power on the axial line is evenly distributed.

Fig. 7 show the comparison between the surface temperature of the axial fuel plate predicted by SACOS-PLATE and the measured value. It is worth mentioning that the power of the two tests is different, 25 kW and 19 kW respectively. From the comparison of the data and the predicted results, it can be seen that at all axial measurement points, the prediction of the surface temperature of the fuel plate by the Colburn correlation is in the best agreement with the experimental results (the maximum error is less than 5 K), and the Mikhayev correlation has the largest difference in the prediction of the measured data (The maximum error is 16.8 K). In addition, most correlations tend to underestimate the surface temperature of the fuel plate. Moreover, when it comes to the average error of different axial heights, the Colburn correlation predicted in the experiment of 19 kW is even smaller, which is 1.7 K different from the experimental result.

The root mean square deviations (RMSD) results presented in Table 4 follow a similar pattern. The RMSD for the Colburn correlation is the smallest, with values of 1.43 K and 2.20 K for the two conditions, respectively. In contrast, the Mikhayev correlation exhibits the largest RMSD, with values of 14.32 K and 16.43 K for the two conditions. Additionally, the RMSD for condition 1 is consistently smaller than that for operating condition 2, indicating that the code performs better in predicting the low-power condition of the RA-6 experiment.

4. Validation of code using JRR-3 test data

4.1. Brief introduction of JRR-3 experiment

JRR3 is a low-enriched uranium silicide plate-fuel research reactor with 26 standard fuel assemblies in the core and its total nominal heating power is 20 MW. This reactor operates in two modes: forced circulation and natural circulation (Albati et al., 2014).

Table 5 shows some geometric parameters of the JRR-3 fuel assembly. Table 6 lists some design parameters under normal and steady-state operating conditions. It is worth mentioning that this research reactor can not only cool the core through coolant circulation under normal operating conditions (20 MW), but also take away the core heat by natural circulation under simulated accident conditions (0.2 MW).

4.2. SACOS-PLATE geometrical model of the JRR-3 test section

In the modeling, each coolant channel is axially divided into 77 nodes (test section 770 mm). The fuel plate is divided into 6 nodes (4 fuels and 2 cladding) in the transverse direction.

Unlike the RA-6 experiment that uses electric heating in Chapter 3, JRR-3 is a research reactor that releases heat through nuclear fission, so its axial power is not uniform. Based on the data in the NEA database and calculations, the axial line power distribution of the nuclear reactor can be obtained as shown in Fig. 8. It's worth mentioning that the power distribution presented in the cited literature is illustrated through dimensionless axial power distribution curves. Due to the likelihood of the total heat transfer area calculated based on the geometric parameters of the fuel components being higher than the actual total heat transfer area, the numerical values of the axial power distribution, derived from the reactor's total power, total heat transfer area, and dimensionless axial power distribution, may be lower than the actual values. Since 77 nodes are divided in the axial direction, the line power for 77 axial nodes is also input in the code.

4.3. Simulation and analysis

Due to lack of relevant experimental data, this study uses the simulation results of the TMAP&COOLOD-N2 code in Literature (Albati et al., 2014) for comparison. In this chapter, SACOS-PLATE utilizes correlations consistent with TMAP&COOLOD-N2. The single-phase heat transfer correlation is the Dittus-Boelter correlation, the single-phase friction resistance coefficient correlation is Blasius correlation and the CHF correlation is the Sudo correlation.

Fig. 9 shows the two-dimensional temperature field distribution of the fuel plate simulated by the code. Different from the calculation results in Chapter 3, due to the use of non-uniformly distributed linear power, the highest temperature of the fuel plate appears above the center. In the horizontal direction, for the same reason, the coolant has the largest temperature rise at the center and the lowest at both ends. Fig. 10 shows the simulation results of the axial fluid temperature, wall temperature and DNBR with TMAP&COOLOD-N2 and SACOS-PLATE codes, respectively.

It can be seen from Fig. 10(a) that the coolant temperature rise predicted by TMAP&COOLOD-N2 is about 8.5 K, while the coolant temperature rise predicted by SACOS-PLATE is about 7 K. This may be because the axial power distribution used by SACOS-PLATE is lower than that of TMAP&COOLOD-N2. In Fig. 10(b), the changing trends of the wall temperature simulated by TMAP&COOLOD-N2 and SACOS-PLATE are almost the same, but the predicted results of SACOS-PLATE show a downward abrupt change at the entrance and exit. This is because the test section (20 mm in total) where the inlet and outlet centerline power are 0 is considered in the modeling of this study. For Fig. 10(c), the DNBR calculated by TMAP&COOLOD-N2 and SACOS-PLATE keeps the same trend along the axial direction, and the minimum DNBR occurs in the position above the axial center. Moreover, the MDNBR predicted by SACOS-PLATE is almost identical to the results predicted by TMAP&COOLOD-N2.

5. Validation of code using the Bettis experimental program data

5.1. Brief introduction of the Bettis experiment

The Bettis experimental program (DeBortoli, 1958) measured thousands of data points for the critical heat flux in a rectangular channel under forced circulation conditions. These data take into account the effects of flow channel geometry, pressure (above 3.45 Mpa), temperature, and mass flow rate. In the Bettis experimental program, satisfactory critical heat flux correlations were not obtained for all variables in all ranges. However, the large number of data points obtained can provide validation for the current numerical simulation studies.

There are two experimental loops (Jacket et al., 1958) in the Bettis Thermal and Hydraulics Laboratory, one for the critical heat flux test and the other for the pressure drop test. Both loops can withstand high temperatures and pressures from 500 psia to 2000 psia.

5.2. SACOS-PLATE geometrical model of the Bettis test section

Different from the previous calculations, the comparison of critical heat flux under different working conditions is emphasized in this chapter, and other parameters are barely considered. In each case, only one critical heat flux was obtained and compared with that in the Bettis experimental program. The relevant design input parameters are shown in Table 7.

The parameters of the cases selected in this study are shown in Table 8. It is worth mentioning that Bettis Laboratory (Troy and Roarty, 1956) has proved through experiments that the critical heat flux is essentially independent of test section material. Therefore, the fuel plate model of fuel and cladding is still used in current study.

5.3. Simulation and analysis

In this study, 10 experimental conditions in the Bettis test were simulated. Sudo correlation is used, as in the previous two chapters, to calculate the critical heat flux. The CHF calculated by the code is compared with the experimental data, as shown in Fig. 11. Most of the simulation results are between 90 % and 110 % of the experimental results, with an average relative error of 7.1 % between the predicted CHF and the experimental ones. This results clearly show that the Sudo correlation has good predictive ability for CHF under rectangular channel, and also prove the computational capability of SACOS-PLATE.

6. Conclusion

XJTU-NuTheL has been committed to developing SACOS series nuclear reactor subchannel analysis codes for many years. In this study, based on the original SACOS code, the flow distribution model, the plate heat conduction model, the heat distribution model and the flow heat transfer model were modified, and the SACOS-PLATE code for subchannel analysis of plate fuel elements was developed. In this present study, the simulation capability of the code was validated by using experimental data from RA-6, and the different flow and heat transfer models added were also compared. The results show that the Colburn correlation can converge to the optimal simulation results effectively for the RA-6 experiment. The maximum axial wall temperature error is 5 K, and the average wall temperature error at different axial heights is 1.7 K. At the same time, it also proved the steady-state computing capability of SACOS-PLATE. In addition, this study also simulated the fuel elements of the JRR-3 research reactor and compared them with the TMAP&COOLOD-N2 code. The results show that the predicted coolant temperature rise between SACOS-PLATE and TMAP&COOLOD-N2 differed by 1.5 K, while the prediction results of MDNBR are also relatively close, which further proves the feasibility and effectiveness of SACOS-PLATE to simulate plate fuel elements. Finally, the data of the Bettis experiment were used to validate the critical heat flux calculation in the code, and the results proved that the error between the predicted value and the experimental value of the Sudo correlation is less than 10 %.

However, the current work is only an innovative startup, and the work of improving and validating the code is far from enough. For the simulation results of different correlations, more in-depth research and analysis remains to be further studied. In the future, SACOS-PLATE will also supplement two-dimensional heat transfer model to realize transient analysis of the flow channels in the first wall of the fusion reactors.

CRediT authorship contribution statement

Ruiyu Sun: Writing – original draft, Writing – review & editing. **Minyang Gui:** Resources. **Jinshun Wang:** Methodology. **Ronghua Chen:** Supervision. **Kui Zhang:** Formal analysis. **Wenxi Tian:** Software. **Suizheng Qiu:** Funding acquisition. **G.H. Su:** Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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